EARTH AND PLANETARY SCIENCE LETTERS 3 (1967) 439-443. NORTH-HOLLAND PUBLISHING COMP., AMSTERDAM

THE MAGNETIZATION OF THE DOTSWOOD RED BEDS (QUEENSLAND)

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Received 21 December 1967

1. INTRODUCTION

The polar wandering curves for most continents for periods older than the Carboniferous are only sketchily known. Yet it is these periods that are of most interest in testing the hypothesis of continental drift.

As part of a program to obtain more palaeomagnetic data for the Devonian of Australia, the magnetization of the red beds within the Dotswood formation was studied. The results did not yield a Devonian pole position, but they are of interest in that they provide another example of remagnetization by viscous partial thermoremanent magnetization of red beds.

The Dotswood formation occurs in the Star Basin, about 17 miles northwest of Mingela in North Queensland (longitude 176°25′ E, latitude 19°45′ S). The base of the formation is formed by a thick sequence of coarse grained felspathic sandstones and greentuff beds. These are followed by red sandstones and purple red shales. To the top the formation becomes conglomeratic. The Dotswood formation overlies the Fannick River group, which on fossil evidence has been assigned to the Givetian. It is overlain by the Myrtlivale Beds which have a Faminian fossil assemblage. The whole overlies the Ravenshoe granodiorite which was dated at 420 m.y. [1]. Between the Dotswood and Fannick River group there is a rather doubtful unconformity. The age of the Dotswood formation is therefore almost certainly upper Devonian and may possibly be Framian.

The red beds are essentially flatlying with dips rarely exceeding 10 degrees. Ripple marks, fossilised mudcracks and crossbedding are readily observed in the field and point to lagoon type depositions.

2. PALAEOMAGNETISM

50 samples were collected through several hundred feet of red beds and grey felspathic sandstones. The exposures were mainly dry creek beds. The intensity of magnetization of the felspathic sandstones is close to the measuring limit of the astatic magnetometer. Although the directions do agree with those from the red beds, no thermal or AC cleaning could be done on them and hence they are not further considered.

The red sandstones are relatively strongly magnetized. The directions of NRM before cleaning are in a distinct group with high positive inclination. The mean (fig. 1) is well away from the present field direction which indicates stability. However, the direction of magnetization does not agree with Devonian direction determined previously for Australia [2, 3]. To test the stability further, six specimens were thermally demagnetized in steps of about 75° to 600°C. In most cases a change in the direction occurred below 250°C, which can be shown to be due to a component of temporary magnetization along the present field. The scatter of the group of six specimens was smallest at about 250°C. At higher temperatures the directions remain essentially constant although the scatter increases somewhat probably due to the increased experimental error, associated with the decreasing intensity. At 600°C the intensity of magnetization was however too low to yield a significant determination of the direction.

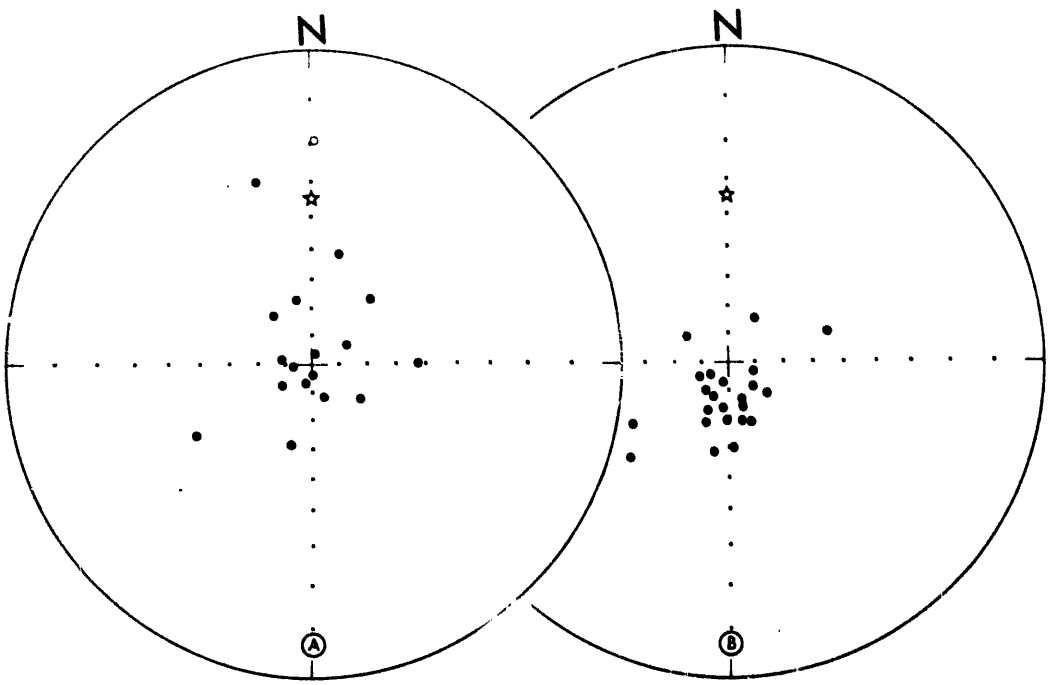


Fig. 1. Directions of magnetization of the Dotswood Red Beds before (A) and after (B). Thermal cleaning to 250°C.

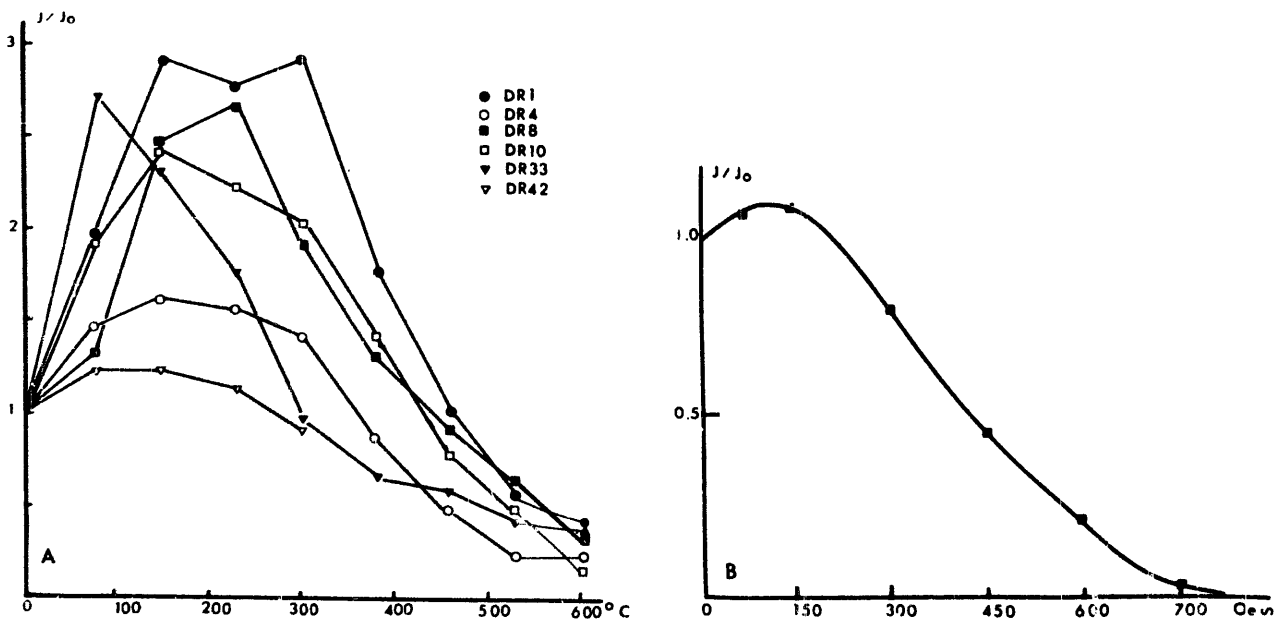


Fig. 2. Thermal (A) and alternating field (B) demagnetization curves for the Dotswood Red Beds.

Two specimens of each sample were then 'cleaned' by heating to 250°C and cooling in zero field. The resulting sample means are shown in fig. 1. It can be seen that the scatter is greatly reduced and tight grouping of the direction resulted. The mean direction is not much different to that of the untreated samples.

The thermal demagnetization curves of the six specimens are shown in fig. 2. The initial increase in the intensity is due to the removal of the temporary component of magnetization which is directed along the present field and opposes the stable NRM direction. At temperatures above 250°C, the intensity falls rapidly to below the measuring limit at about 600°C.

One specimen was also demagnetized by alternating magnetic fields. The change in directions was the same as in the thermal demagnetization experiment. The demagnetization curve, fig. 2, is noteworthy in that it shows an, for red beds, unusual rapid decrease in intensity with increasing AC field.

The AC and thermal demagnetization characteristics of these rocks stronly suggest that magnetite rather than hematite is the carrier of the magnetization. To confirm this a number of specimens were polished and examined in reflected light. Most of the opaque material is very fine grained and difficult to identify with certainty. The larger grains however are clearly martite and have remnants of magnetite in them. These remnants sometimes make up as much as 30% of the martite grain.

It seems reasonable therefore to attribute the NRM to these magnetite remnants, the contribution of the hematite being too small to observe.

3. DISCUSSION

The pole position and statistical parameters [4] for the Dotswood red beds are given in table 1. In the absence of a fold or conglomerate test [5], the age of the magnetization cannot be established with certainty. However, the following arguments suggest that the red beds became remagnetized probably in the Upper Carboniferous.

Firstly, the Dotswood pole disagrees with Devonian poles previously [2, 3] determined for Australia but since these poles are based on data not treated by demagnetization procedures, this discrepancy may not be significant. More important however, is the fact that the Dotswood pole disagrees with the poles of the Mugga Porphyry and the lower Kuttung Volcanics, which are Upper Silurian and Lower Carboniferous respectively. These two poles are based on treated data and the remarkable agreement between them was interpreted by Irving [6] to indicate that little polar movement occurred during the Devonian and Lower Carboniferous for Australia. Irving further suggested that although the pole moved rapidly during the Carboniferous, it remained more or less stationary throughout the Upper Palaeozoic and Mesozoic. The Dotswood pole agrees remarkably well with these Upper Palaeozoic and Mesozoic poles, suggesting remagnetization in that time interval.

Other evidence pointing towards remagnetization is the unusual tight grouping of the directions of magnetization and the fact that the palaeolatitude of these Dotswood red beds is much higher than would be ex-

Table 1

Directions of magnetization and statistical parameter for the Dotswood Red Beds and Percy Creek volcanics. N = number of samples, R = length of resultant vector, K = dispersion factor, α = circle of confidence at 95% probability level, D and I mean declination and inclination of remanent vector.

Formation	Geological age	Statistical parameters				Direction		Pole coordinates		Treatment
		N	R	K	α_{95}	D	I			
Dotswood Red Beds	Upper Devonian	19	17.96	17.39	8.2	195.1	+75.1	135.6E	−46.1S	250°C thermal
Percy Creek volcanics	Late middle upper carboniferous	5	4.80	19.66	18.2	244.0	+85.4	137.2E	−23.3S	300 oersted AF

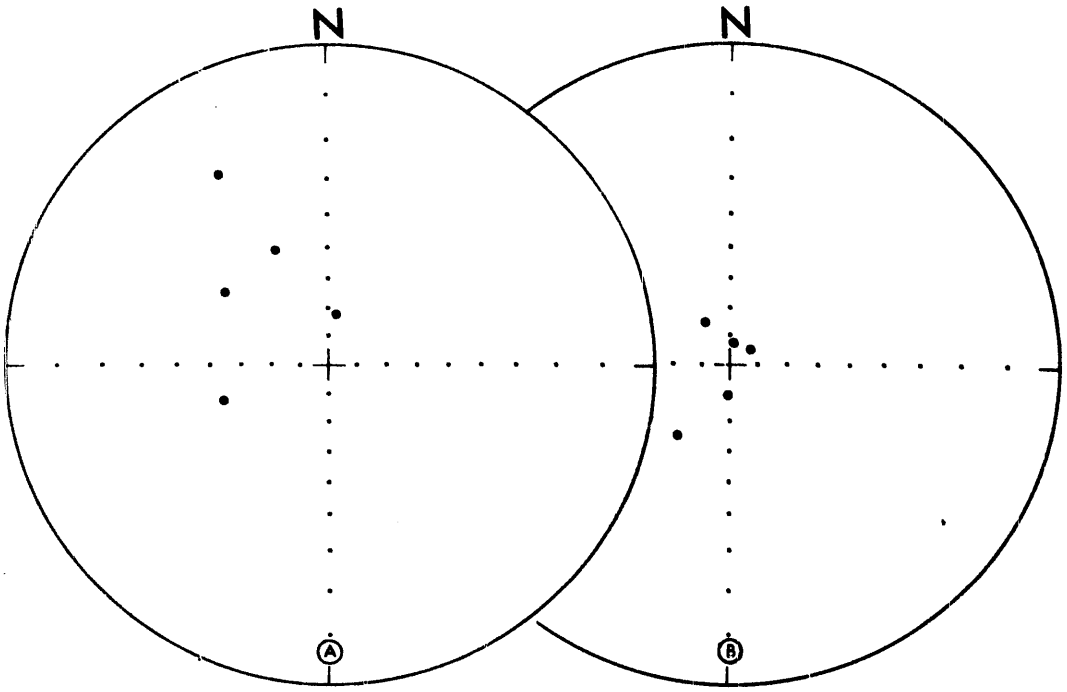


Fig. 3. Directions of magnetization of the Percy Creek volcanics before (A) and after (B) alternating field demagnetization.

pected from the present day distribution of red beds or from the palaeolatitude spectra for red beds of all geological ages [7]. The main reason for suspecting remagnetization is, however, the fact that the magnetization is carried by magnetite and that the principle blocking temperatures lie between 200°C and 400°C.

The author has previously [8, 9] reported results from thermal demagnetization experiments in the Old Red Sandstones of the Anglo-Welsh Cuvette. It was shown that the dominant component of magnetization was carried by magnetite and imposed on the rocks in the Permian after folding. This stable but secondary component could only be removed after heating to 600°C. To explain the remagnetization it was suggested that during deep burial the relaxation time had decreased sufficiently to cause a change in the direction of magnetization which was then subsequently frozen in during uplift. The mechanism was termed viscous P.T.R.M. Recent laboratory experiments by Briden [10] have shown such a mechanism to be feasible. It is expected to be mainly operative in magnetite bearing rocks. Compared with the Old Red Sandstones, the blocking temperatures of the Dotswood red beds are somewhat lower and would make them therefore more susceptible to remagnetization.

The two most obvious geological processes that could be responsible for V P.T.R.M., are deep burial or regional heating for example by volcanic activity. To test this possibility, five oriented samples form the Percy Creek volcanics occurring in the same area were collected by D. Wyatt. The volcanics unconformably overlie middle Carboniferous sediments and are thought to be late Middle or Upper Carboniferous in age. In fig. 3 the directions of magnetization are shown with respect to the bedding plane before and after treatment in an alternating field of 300 oersted (peak). This field was used because it resulted in the smallest within sample dispersion. The pole and polarity of the Percy Creek volcanics agrees with other Kiaman (Upper Palaeozoic) poles determined for Australia [6]. The results are summarized in table 1.

Comparison of fig. 1 and fig. 3 shows clearly that the magnetization of the Dotswood red beds is identical to that of the Percy Creek volcanics, both in direction and polarity. It is therefore reasonable to suggest that the Dotswood red beds became remagnetized during the Upper Carboniferous as the result of regional heating associated with Permo-Carboniferous volcanic activity.

ACKNOWLEDGEMENT

The author is grateful to D. T. Wyatt who introduced him to the geology of the Star basins and accompanied him on the field trip. The geological description of the area in this paper is mainly due to him.

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